

Based on scientific

Gravitational force is because & mass more the mass more will be the gravitational force.

Gravitational force is centre seeking force i.e. centre of mass of one object attracts centre of mass of other.

Therefore any object around the Earth falls vertically downwards.

Circular motion & Centripetal force

1) Motion of the particle along the circumference of circle is called circular motion.

2) Centripetal force:- Force responsible for circular motion is called centripetal force.

Kepler's laws of planetary motion

1) The orbit of the planet is an ellipse with the sun at one of the foci.

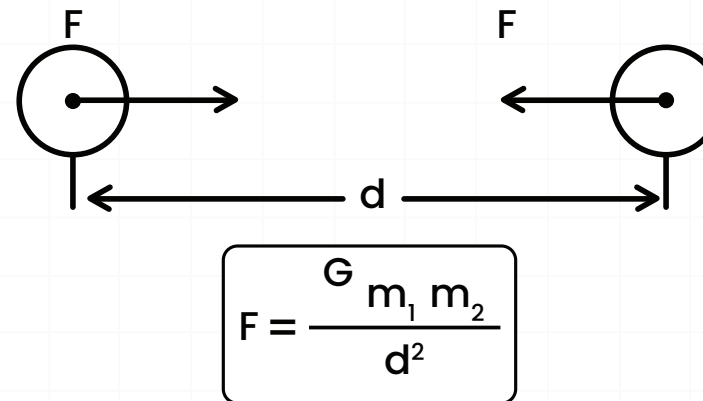
2) The line joining the planet and the sun sweeps equal area in equal intervals of time.

3) The square of the time period of revolution around the sun is directly proportional to the cube of the mean distance of a planet from the sun.

$$T^2 \propto r^3 \text{ i.e. } \frac{T^2}{r^3} = \text{constant}$$

Newton's Universal law of gravitation

Every object in the universe attract every other object which is directly proportional to the product of the masses and inversely proportional to the square of the distance between them.



Earths gravitational force

Moving objects around the earth experience change in its velocity.

Change in velocity means acceleration

According to Newton's and law of motion acceleration occurs due to the force acting on it. This force acts towards the centre of the earth.

Earths gravitational acceleration

Formula of earth's gravitational acceleration

$$g = \frac{GM}{R^2}$$

1. Value of acceleration due to gravity is maximum on the earth's surface it decreases with height and depth. Mass & weight

[Mass: The amount of matter present in an object is called mass
& Weight is the foree acting on that mass due to earths]

GRAVITATION

Free fall

Motion of an object under gravity only is called free fall
Changes in equations of motion for free fall are as follows.

$$V = u + at \rightarrow v = gt$$

$$s = ut + \frac{1}{2} at^2 \rightarrow h = \frac{1}{2} gt^2$$

$$v^2 = u^2 + 2as \rightarrow v^2 = 2gt$$

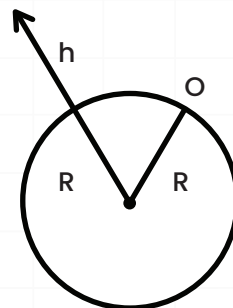
Gravitational Potential Energy

As you know the formula of potential energy $P.E. = mgh$
Actually it should be

$$P.E. = mg(R + h)$$

On the surface of the earth

$$P.E. = mgR$$



As we know

$$g = \frac{GM}{R^2}$$

$$P.E. = m \times \frac{GM}{R^2} \times R = \frac{GMm}{R}$$

As this energy stored in the body it is consider as -ve

Escape Velocity

The minimum velocity with which an object should be launched to escape from the earth's gravitational field is called escape velocity.

$$V_{esc} = \sqrt{\frac{2GM}{R}}$$

Periodic classification of elements



Traids

Group of three elements each, having similar chemical properties and called them triads.

Dobereiner's Triad

The three elements in a triad in an increasing order of atomic mass and showed that the atomic mass of the middle element was approximately equal to the mean of the atomic masses of the other two elements.

Sr. No.	Triad	Element-1 Actual atomic mass(a)	Element-2		Element-3 Actual atomic mass (c)
			Mean = $\frac{a+c}{2}$	Actual atomic mass	
1	Li, Na, K	Lithium (Li) 6.9	Sodium $\frac{6.9+39.1}{2} = 23.0$	(Na) 23.0	Potassium (K) 39.1
2	Ca, Sr, Ba	Calcium (Ca) 40.1	Strontium $\frac{40.1+137.3}{2} = 88.7$	(Sr) 87.6	Barium (Ba) 137.3
3	Cl, Br, I	Chlorine (Cl) 35.5	Bromine $\frac{35.5+126.9}{2} = 81.2$	(Br) 79.9	Iodine (I) 126.9

Newlands' Law of Octaves

Newlands arranged the elements known at that time in an increasing order of their atomic masses. Newlands found that every eighth element had properties similar to those of the first... He called the similarity observed in the eighth and the first element as the Law of octaves

Mendeleev's Periodic table law

Properties of elements are periodic function of their atomic masses.

Periodic classification of elements

Merits of Mendeleev's periodic table

Atomic masses of some elements corrected

Mendeleev kept vacant places in the periodic table for elements not discovered

Mendeleev created the zero' group without disturbing the original periodic table

Merits of Mendeleev's periodic table

1. Position of isotopes

2. Position of Hydrogen

3. Position of Cobalt and Nickel

Merits of Mendeleev's periodic table

Properties of elements are a periodic function of their atomic numbers

Modern periodic table: long form of the periodic table

Atomic no. No of protons No of electrons

Structure of the Modern Periodic Table

7 periods (rows)

18 Groups (Column)

Categorization

(1) (i) Metals
(ii) Non metals
(iii) Metalloid
(iv) Noble gases

(2) (i) s block
(ii) p block
(iii) d block
(iv) f block

(3) (i) s block
(ii) Transition elements
(iii) Inner transition elements
(iv) Noble gas

Modern periodic Table and electronic Configuration of Elements

Within a period, the neighbouring elements differ slightly in their properties while distant elements differ widely in their properties.

Groups and electronic configuration

Number of valence electrons of all the elements in group 1 & 2 and from 13-18 will be same

Periods and electronic configuration

No of shells = Period No

Ex. Na11 (2,8,1) belongs to period no 3

PERIODIC TRENDS IN THE MODERN PERIODIC TABLE

Atomic size (radius)

INCREASE

left to right in a period

DECREASE

Top to bottom in a group

Metallic Character

INCREASE

left to right in a period

INCREASE

Top to bottom in a group

Non-metallic character

INCREASE

left to right in a period

DECREASE

Top to bottom in a group

Electro negativity

INCREASE

left to right in a period

DECREASE

Top to bottom in a group

Metallic Character

DECREASE

left to right in a period

INCREASE

Top to bottom in a group



CHEMICAL REACTIONS AND EQUATIONS

PHYSICAL CHANGE

- No new substances formed.
- Change of shape, size, color or density of original substance

Eg:

- Boiling of water → water vapour
- Melting of ice → water
- Dissolution of salt into water

CHEMICAL CHANGE

- New substances formed.
- Change in chemical properties

Eg:

- Rusting of iron
- Curding of milk
- Candle Burning (wax combines with atmospheric O_2 to give CO_2)
- Combustion of LPG

CHEMICAL REACTION

- One or more substances react to form new compounds with new properties.

Eg:

- Heating copper wire in presence of air.
- Magnesium ribbon is usually cleaned before burning in air as it has a coating of magnesium oxide due to a chemical reaction this hinders the process of burning.

EFFECTS OF OXIDATION (In Everyday life)

REPRESENTATION

RANCIDITY

- Oxidation of oils and fats in food resulting in bad smell and taste.

Eg:

- By using antioxidants (reducing agents)
- By packing in airtight containers
- By keeping vegetables, fruits in fridge
- By flushing (ex. bag of chips) with N_2 gas

CORROSION

- The process of slow destruction of metals by atmospheric gases, water vapour etc. so as to form a layer of oxide on their surface.

How to prevent
Painting, Greasing, Galvanisation, Electroplating.

COMBINATION REACTION

- It is a symbolic representation of chemical reaction through symbols and formulae.
- Physical state of an element is written as gas(g), liquid(l), solid(s) and when solution made in water called aqueous (aq).

The reaction conditions such as temperature, pressure and catalyst are indicated above or below the arrow.

BALANCING CHEMICAL EQUATION

- Number of atoms of each element should be equal on both sides.
- Follows law of conservation of mass (Mass Reactant = Mass Products).

Eg: $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$

WRITING CHEMICAL EQUATION

- Skeletal equation: number of atoms of each element on both sides are not equal.
- Balanced chemical equation: number of atoms of each element on both sides are equal. $2H_2 + O_2 \rightarrow 2H_2O$

TYPES

DISPLACEMENT REACTION

- More reactive element displaces less reactive

Eg:

$$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$$

(Fe is more reactive than Cu) Fe changes $CuSO_4$ color to light green $FeSO_4$

OXIDATION REACTION

- Gain of oxygen atoms.
- Loss of hydrogen atoms.

Eg: $2Cu(s) + O_2 \rightarrow 2CuO$

REDUCTION REACTION

- Gain of hydrogen atoms.
- Loss of oxygen atoms.

Eg: $CuO + H_2 \rightarrow Cu + H_2O$

DECOMPOSITION REACTION

- One compound (or reactant) requires energy (heat, light or electricity) to break into two or more elements/compounds or products.

Eg: $FeSO_4(s) \xrightarrow{\text{heat}} Fe_2O_3 + SO_2 + SO_3$

Decomposition reactions are always endothermic

DOUBLE DISPLACEMENT REACTION

- Two compounds exchange their ions to form new compounds.

Eg: $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$

After a chemical reaction when one of the products formed in solid state called precipitation.

Eg: $Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$

- Takes place only when there is a formation of highly soluble salt.

COMBINATION REACTION

- Two element or compound combine to form new compound.

Eg:

- $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$
- $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2 + \text{Heat}$ (Used in whitewashing)

EXOTHERMIC REACTION

- Heat is Produced Or Released.

Eg:

- Respiration during digestion CO_2 in food breaks down to glucose this glucose with O_2 releases energy.
- $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O$

OXIDISING AGENT

- Substance which gives oxygen or gains hydrogen or, that which reduces itself is an Oxidising agent.

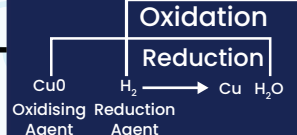
REDUCING AGENT

- Substance which gives hydrogen or gains oxygen or, that which oxidises itself is an reducing agent.

REDOX REACTION

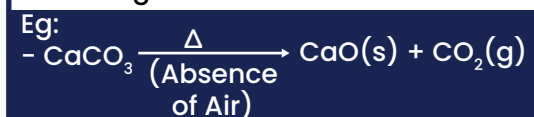
- Reaction in which both oxidation and reduction takes place

Eg: $ZnO + C \rightarrow Zn + CO$



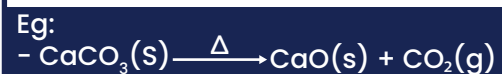
THERMAL DECOMPOSITION

- A single compound breaks into two or more elements/ compounds on heating.



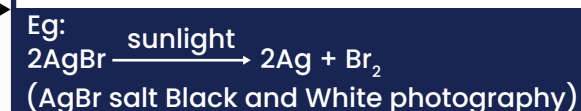
Endothermic Reaction

- Heat is absorbed.



PHOTOCHEMICAL DECOMPOSITION

- A reactant breaks down in presence of sunlight.



ELECTROLYTIC DECOMPOSITION

- A reactant breaks down in presence of electric current.

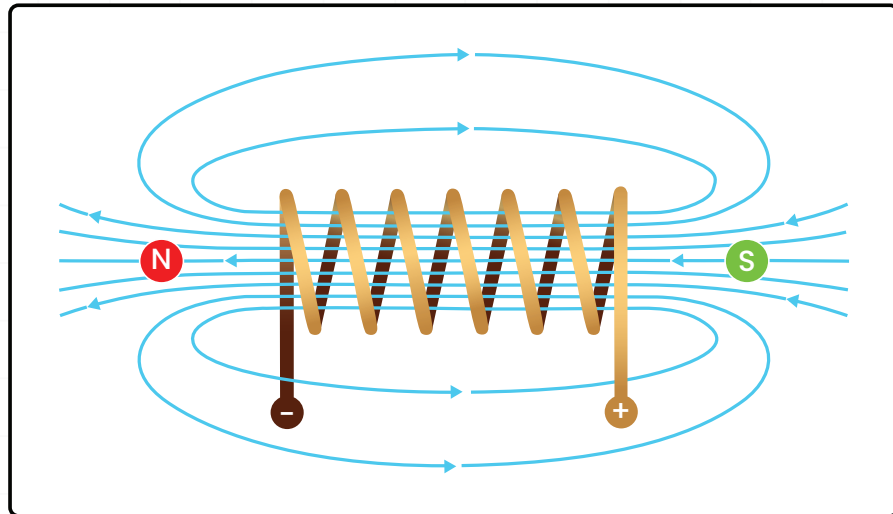


Effects of electric current

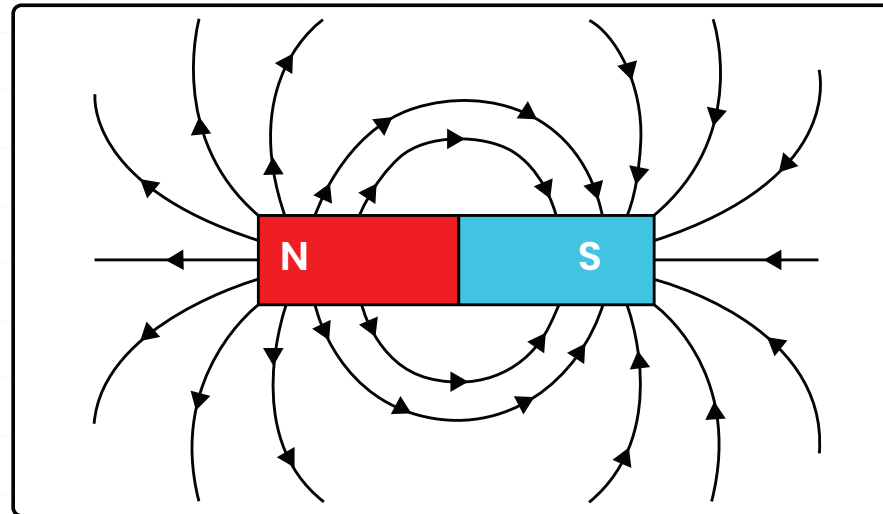
1. Circular motion & Centripetal force

- Representation of the magnetic field path along which an imaginary free north pole would tend to move.
- The tangent to any point on the magnetic field lines gives the direction of the magnetic field at that point.
- Field Lines North to South pole.

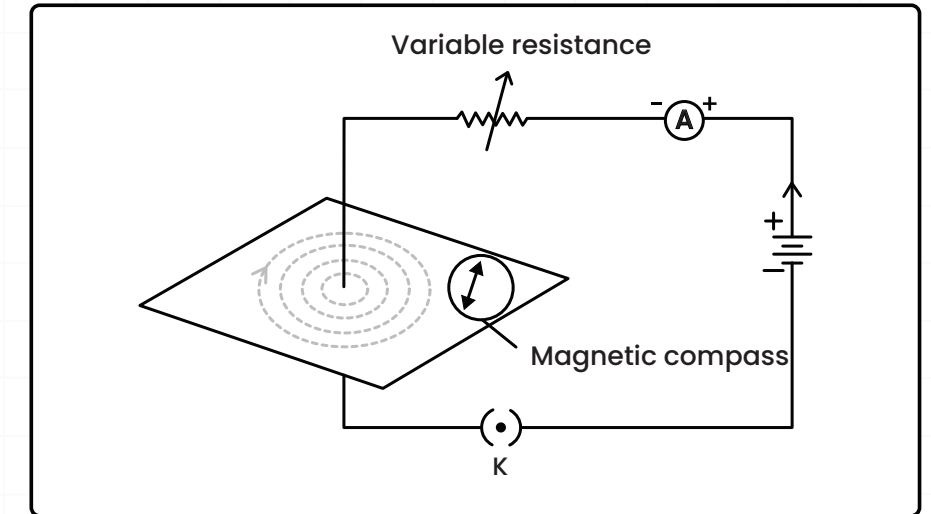
2. Magnetic Fields



1.) Field lines around a solenoid

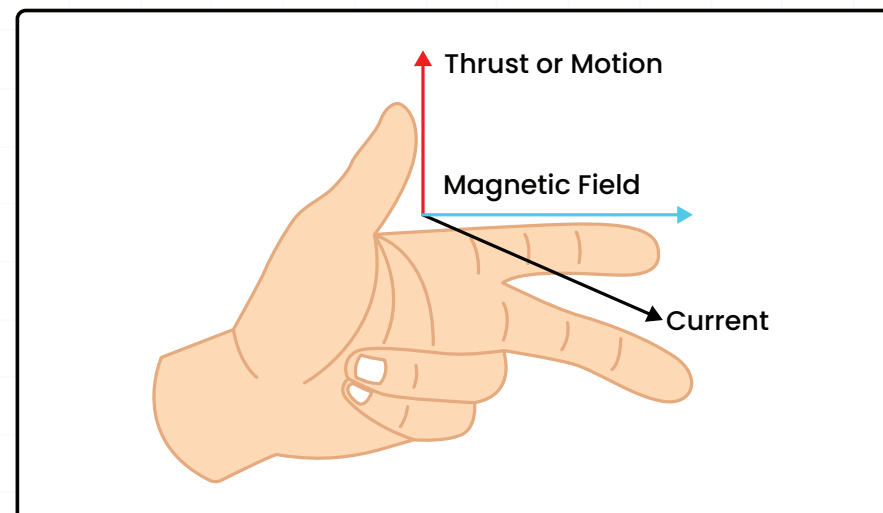


2.) Field lines around a bar magnet



3.) Field lines around a straight conducting wire

3. Fleming's Left Hand Rule



4. Electric Motor

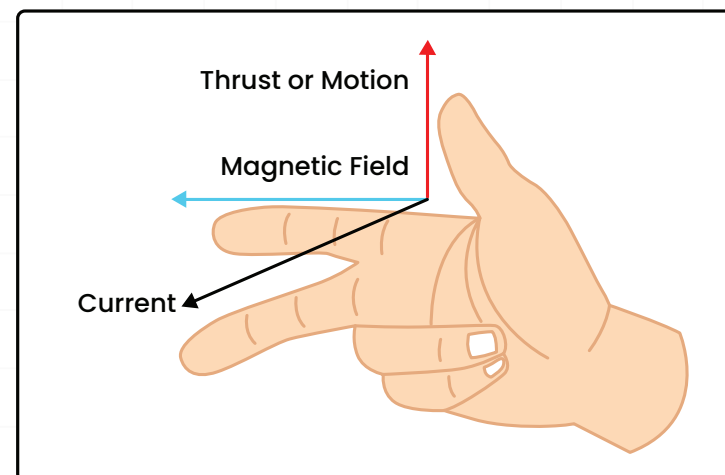
- Converts electric energy into mechanical energy
- Works on the principal of Fleming's left hand rule

5. Electromagnetic Induction

Proces by which a change in magnetic field induces a current in conductor

6. Fleming's Right Hand Rule

Proces by which a change in magnetic field induces a current in conductor



7. Electric Generator

- Converts mechanical energy into eletric energy
- Work on the basis of electromagnetic induction (Fleming's Right hand rule)

6. Domestic Electric Current

- Earth wire (with green insulation cover) safety measures to take care of leakage.
- Supply electricity to circuit withing home

(a) Neutral wire (Negative)

(b) Live wire (positive)

7. Fuse

- Protects the appliances from short circulating (overloading).



8. Heating Effect of Electric Current

- Joule's law of heating

The amount of heat produced in a conductor is proportional to

1. Square of current flowing through the conductor.
2. Resistance of the conductor.
3. Time for which current is passed

$$P = \frac{V^2}{R} \text{ (unit is watt)}$$

- Electric power
- Electric energy = $VIt = I^2Rt$

Heat

Latent Heat

Latent Heat of fusion

When solid to liquid or liquid to solid conversion will be there latent heat of fusion can be experienced.

Latent heat of vaporization

When liquid to gas or gas to liquid conversion will be there latent heat of vaporization can be experienced.

Regelation

When pressure applied on ice melting point/ freezing point temperature drops. When pressure is removed melting point/freezing point temperature of ice restores to zero.

Anomalous Behaviour of Water

Normal Behaviour of water

Temp • Density • Volume • Expansion
Temp • Density Volume Contraction

Anomalous Behaviour of water

Temp • Density • Volume • Expansion
Temp • Density Volume Contraction

Saturated Air

When air contains maximum possible water vapour it is said to be saturated air.

Unsaturated Air

If vapour contained in air is less than that the maximum limit, it is said to be unsaturated air

Dew point temperature

Unsaturated air at a certain temperature is taken and its temperature is reached at which the air becomes saturated with vapour. This temperature is called as dew point temperature.

Absolute Humidity

The mass of vapour present in a unit volume of air is called absolute humidity.

% Relative Humidity = $\frac{\text{Absolute Humidity}}{\text{Saturation Vapour Pressure}} \times 100$

Unit of Heat

EMBED Equation. DSMT4

Specific Heat capacity

The amount of heat energy required to raise the temperature of a unit mass of an object by 1°C is called the sp. heat of that object.

Heat exchange principal

Heat energy lost by the hot object = Heat energy gained by the cold object.

— Refraction of Light —

1. Laws of Refraction

- (i) The incident ray, normal & refracted ray lie in one plane.
- (ii) The ratio of sin of angle of incidence and sin of angle of refraction is always constant for given pair of media.

2. Refractive Index

$$(i) \quad n = \frac{\sin i}{\sin r}$$

$$(ii) \quad {}^1n_2 = \frac{\text{Velocity of light in medium 1}}{\text{Velocity of light in medium 2}}$$

3. Twinkling of Stars

It is an optical phenomenon caused by the total internal reflection of light from distant objects.

4. Apparent Position of a Star

Since the atmosphere bends the starlight towards normal, the apparent position of the star appears to be slightly different from actual position.

5. Dispersion of Light

The process of separation of light into its component colours while passing through a medium is called the dispersion of light.

6. Partial and total internal Reflection

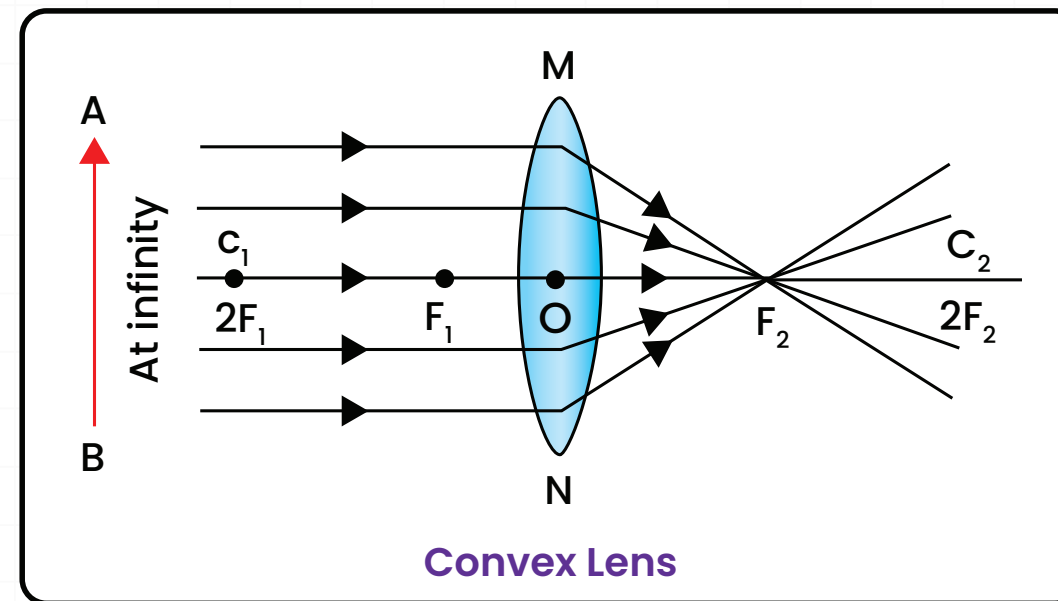
1. When a light enters a rarer medium from a denser medium, it gets partially reflected this is called partial reflection.
2. From a particular value of i , the value of r becomes equal to 90° . This value of i is called the critical angle.
3. For angle of incidence larger than the critical angle, the angle of refraction is larger than 90° . Such rays return to the denser medium. This is called total internal reflection.

7. Rainbow

1. Rainbow is a beautiful natural phenomenon
2. It is a combined effect of dispersion, refraction and total reflection.

— Lenses —

2. Convex Lens Converging



3. Position of the object

At Infinity	Beyond $2F_1$	At $2F_1$	Between $2F_1$ and F_1	At F_1	Between F_1 and optical centre
Image at F_2 (real)	Image between F_2 and $2F_2$ (real)	Image at $2F_2$ (real)	Image beyond $2F_2$ (real)	Image at infinity	Image on the same side of the lens at the object.

4. Uses

- Simple Microscope
- Compound Microscope
- Telescope
- Camera Spectrometer
- Spectacles

5. Refraction of Light

Snells Law

The ratio of the sine of angle of incidence to the sine of angle of refraction is always constant $n = \frac{\sin i}{\sin r}$

The incident ray, the normal and the refracted ray at the point of incidence all lie in the same plane for the two given transparent medium

6. Refractive Index

$$n = \frac{\text{Velocity of light in medium 1}}{\text{Velocity of light in medium 2}}$$

7. Magnification

Mirrors

$$m = v/u = h_1/h_0$$

8. Formula for Focal Length

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

9. Power of lens

Power is measured in Diopters (D) $P = 1/f$ (in meters)

Concave Lens

Diverging

Uses

- Eye Glasses and contact lenses
- Flashlights
- Peepholes



Metallurgy

Physical Properties of Metals

- Solid state
- Lustre
- Ductility
- Malleability
- Good conductors of heat and electricity
- Hard
- High melting and boiling points.
- Sonorous

Physical Properties of Nonmetals

- Solid, liquid, gaseous state
- Non lustrous
- Non ductile
- Non malleable
- Bad conductors of heat and electricity
- Soft
- Low melting point and boiling point
- Non sonorous



Chemical properties of metals

- Reaction of metals with oxygen Metals combine with oxygen on heating in air and metal oxides are formed.

B. Reaction of metals with water

- Sodium and potassium metal react rapidly and vigorously with water and liberates hydrogen gas.
- Calcium reacts with water slowly and less vigorously.
- Aluminium, iron and zinc do not react with cold or hot water, but they react with steam to form their oxides.

C. Reaction of metals with acids

- Aluminium, magnesium, iron or zinc are treated with dilute sulphuric or hydrochloric acid, sulphate or chloride salts of metals are formed.

D. Reaction of metals with nitric acid

- Nitrate salts of metals are formed on reaction of metals with nitric acid.

Aqua Regia: 3:1 mixture of concentrated hydrochloric acid and concentrated nitric

E. Reaction of metals with salts of other metals

Metal A + Salt solution of metal B Salt solution of metal A + Metal B. If a metal A displaces another metal B from the solution of its salt then it means that the metal A is more reactive than the metal B.

F. Reaction of metals with nonmetals

Metals do this by losing electrons while nonmetals do this by gaining electrons.



Chemical properties of nonmetals

A. Reaction of nonmetals with oxygen:

- Generally, nonmetals combine with oxygen to form acidic oxides. In some cases, neutral oxides are formed.

B. Reaction of nonmetals with water

- Generally, nonmetals do not react with water, except the halogens.

C. Reaction of dilute acids with nonmetals:

- Generally, nonmetals do not react with dilute acids, halogens are exception to this.

D. Reaction of nonmetals with hydrogen

- Nonmetals react with hydrogen under certain condition (such as proper temperature, pressure, use of catalyst, etc.)



Ionic compounds

The compounds formed from two units, namely cation and anion are called ionic compounds

Properties of ionic compounds

- Solid and hard
- Brittle
- High intermolecular force of attraction
- Water soluble.
- Bad conductor of electricity in solid state.
- Good conductor of electricity in molten or aqueous solutions



Metallurgy

The science and technology regarding the extraction of metals from ores and their purification for the use is called metallurgy.

Occurrence of Metals

Occur in nature in free state but also but are found in combined state as their salts such as oxides, carbonates, sulphides and nitrates.

Basic principles of metallurgy

• Concentration of ores

- A. Separation based on gravitation
 - i. Wilfley table method
 - ii. Hydraulic separation method
- B. Magnetic separation Method
- C. Froth floatation method
- D. Leaching

• Extraction of metals

- A. Extraction of reactive metals
- B. Extraction of moderately reactive metals
- C. Extraction of less reactive metals

• Refining of metals

Corrosion of metals

Prevention of corrosion

1. Galvanizing

2. Tinning

3. Anodization

4. Electroplating

5. Alloying



Carbon and Compounds

Bonds in Carbon compounds

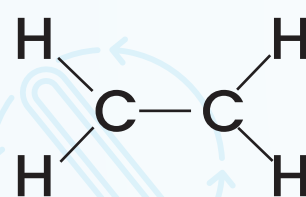
The chemical bond formed by sharing of two valence electron between the two atoms is called covalent bond.

A covalent bond is represented clearly by drawing an electron – dot structure.

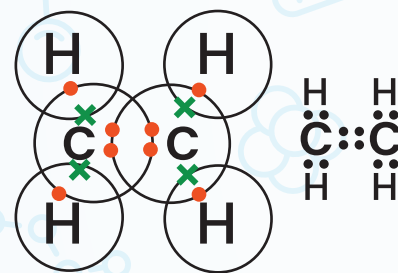
Carbon: A Versatile Element

1) Catenation

2) Tetravalent



Line structure/ structural formula



Electron-dot structures of ethane



Carbon and Compounds

The compounds which contain carbon and hydrogen as the only two elements are called hydrocarbons.



(Hydrocarbon)

Aliphatic (open chain compound)

Saturated Hydrocarbon

Alkene (C_nH_{2n+2})

Unsaturated Hydrocarbon

Alkyne (C_nH_{2n})

Alkynes (C_nH_{2n+2})

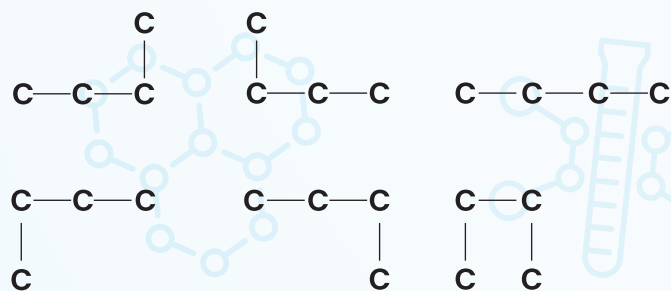
Cyclic (Closed chain compounds)

Alicyclic

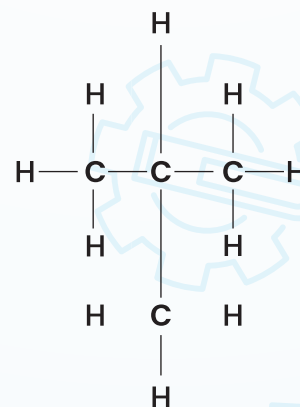
Aromatic



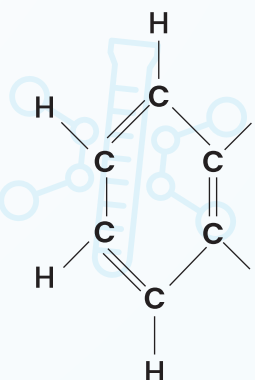
Straight chains, Branched chains and Rings of Carbon atoms



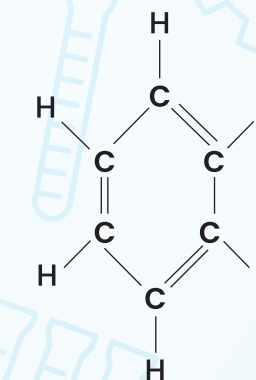
Straight Chain(open)



Branched Chain(open)



Ring (closed)chain



Carbon and compounds



Structural isomerism

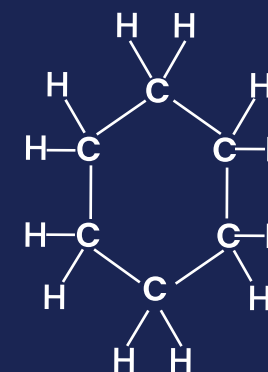
The phenomenon in which compounds having different structural formulae have the same molecular formula is called 'structural isomerism'.

A. The carbon ring in cyclohexane



Ring Structure of Cyclohexane

B. Structural formula of cyclohexane



Functional Groups in Carbon Compounds

Hetero Atom	Functional Group		
	Name	Structural formula	Condensed Structural formula
Halogen (chlorine, bromine, iodine)	Halo (chloro/bromo/iodo)	-X (-Cl, -Br, -I)	-X (-Cl, -Br, -I)
Oxygen	1. Alcohol	-O-H	-OH
	2. Aldehyde	$\begin{array}{c} \text{O} \\ \parallel \\ \text{-C-H} \end{array}$	-CHO
	3. Ketone	$\begin{array}{c} \text{O} \\ \parallel \\ \text{-C-} \end{array}$	-CO-
	4. Carboxylic Acid	$\begin{array}{c} \text{O} \\ \parallel \\ \text{-C-O-H} \end{array}$	-COOH
	5. Ether	-O-	-O-
	6. Ester	$\begin{array}{c} \text{O} \\ \parallel \\ \text{-C-O-} \end{array}$	-COO-
Nitrogen	Amines	$\begin{array}{c} \text{-N-H} \\ \\ \text{H} \end{array}$	-NH ₂

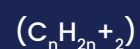
Carbon and compounds



Homologous series

Homologous series is a group of organic compounds having a similar structure and similar chemical properties in which the successive compounds differ by a CH_2 Group.

Homologous series of Alkane of Alkyne

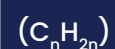


Methane CH_4

Ethane C_2H_6

Propane C_3H_8

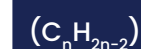
Homologous series of Alkene



Ethene C_2H_4

Propene C_3H_6

Homologous series of Alkyne



Ethyne C_2H_2

Propyne C_3H_4

IUPAC nomenclature system

Halide + Substituents

Parent chain (Main chain)
(Count no. of carbon)

+ Single/double/ triple bond

+Functional group

Chemical Properties of Carbon Compounds

- 1. Combustion:** carbon in the form of various allotropes on ignition in presence of oxygen undergoes combustion to emit heat and light.
- 2. Oxidation:** Carbon compounds start burning by combining easily with oxygen in the air when ignited in air
- 3. Addition Reaction:** When a carbon compound combines with another compound to form a product that contain all the atoms in both the reactants, it is called an addition reaction. Unsaturated.
- 4. Substitution reaction:** The reaction in which the place of one type of atom / group in a reactant is taken by another atom / group of atoms, is called substitution reaction.

Carbon and compounds

Important carbon compounds: Ethanol and Ethanoic Acid Chemical properties of ethanol

(i) Reaction with sodium

(ii) Dehydration reaction

Chemical Properties of Ethanoic Acid

(i) Reaction with base

a. A reaction with strong base

(ii) Esterification Reaction:

Macro molecules and Polymers

1. **Macromolecules**: The giant carbon molecules formed from hundreds of thousands of atoms are called macromolecules.
2. **Natural macromolecules**: The natural macromolecules namely, polysaccharides, proteins and nucleic acids are the supporting pillars of the living world.
3. **Manmade macromolecules**: Manmade fibres wool and silk; elastomers
4. **Polymers**: A macromolecule formed by regular repetition of a small unit is called polymer.
5. **Monomer**: The small unit that repeats regularly to form a polymer is called monomer.

— Space Mission —

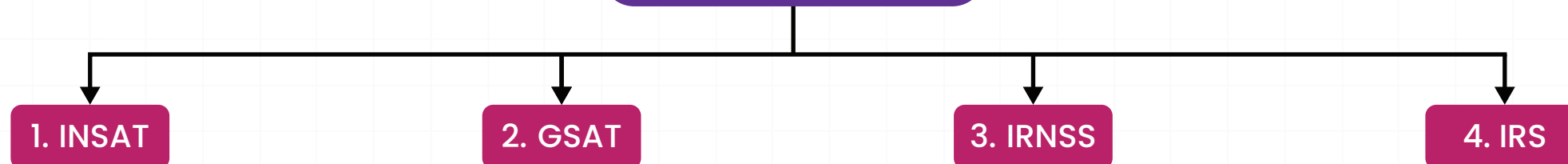
1. Objectives

1. To study earth
2. To study planets and other celestial bodies.

2. Artificial Satellite

Man made object that revolve around the celestial body.

3. Indian Satellites



SLVS



4. Orbital Velocity

$$V_c = \sqrt{\frac{GM}{R+h}}$$

5. Types of Orbit

1. High earth orbit (> 35780 km)
2. Medium earth orbit (2000 km to 35780 km)
3. Low earth orbit (180 km to 2000 km)

6. Moon Mission

1. Luna 2 (1859) – Russia
2. 1962 – 1972 → Moon missions by America
3. 1969 – Neil Arm Strong (First human to step on moon)
3. 1969 – Neil Arm Strong (First human to step on moon)
4. 2008 (ISRO) (Chandrayaan-1)
5. India was the first country to discover water on the moon.

7. Mar's Mission

1. Mass mission is difficult and almost half the missions were unsuccessful
2. ISRO's Mangalyaan launched in Nov. 2013 and was placed into orbit around the mass in September 2014.

8. India and Space Technology

1. Two types of vehicle PSLV and GSLV
2. INSAT and GSAT series working in the field of telecommunication, television broad casting meteorological services.
3. EDVSAT in the field of education.
4. IRS natural resources and its monitoring.
5. IRNSS satellite series (For Navigation)